

1. EXECUTIVE SUMMARY AND CASE CONTEXT

Modern automotive powertrains demand rigorous **engineering reliability**. In this illustrative engineering evaluation, an automotive sub assembly exhibits functional instabilities across its operational lifecycle, manifesting as premature infant mortality failures, random mid life operational breakdowns, and wear out degradation.

The core objective of this **Quality Engineering study** is to establish a robust analytical framework to identify failure modes, quantify risk priorities, model time to failure distributions, and execute systematic root cause mitigation.

2. FMEA METHODOLOGY AND RISK PRIORITISATION

Failure Mode and Effects Analysis (FMEA) serves as the foundational engineering tool to proactively evaluate potential vulnerabilities before operational deployment. The risk matrix evaluates three core dimensions:

- **Severity (S):** Assessment of the clinical and operational consequence of a failure mode on system performance.
- **Occurrence (O):** Statistical frequency or probability that a specific failure cause will manifest.
- **Detection (D):** Capability of current quality controls to intercept the failure mode prior to customer impact.

$$\text{Risk Priority Number (RPN)} = \text{Severity (S)} \times \text{Occurrence (O)} \times \text{Detection (D)}$$

RPN dictates engineering resource allocation for risk mitigation. (Note: Numerical RPN thresholds are maintained per internal standard operating procedures and omitted from this illustrative framework).

3. ILLUSTRATIVE FAILURE MODE ANALYSIS

Failure Mode: Structural or functional degradation under dynamic vehicular loading conditions.

Potential Effect: Sub optimal torque transmission and accelerated assembly mechanical wear.

Potential Investigation Vectors (Non proprietary):

- Component dimensional and metallurgical variance
- Cyclic thermal and mechanical overload conditions
- Lubrication degradation and boundary friction

4. LIMITATIONS OF FMEA IN LIFECYCLE ANALYSIS

While FMEA successfully establishes static risk hierarchies, it operates independently of temporal dynamics. It does not answer a critical reliability engineering question: *“When failures manifest in the field, what chronological pattern do they follow over operational hours?”* Bridging this operational gap requires transitioning from static risk scoring to stochastic reliability modelling via **Weibull Analysis**.

FMEA for Static Risk Identification | Weibull Analysis for Dynamic Lifecycle Behaviour

5. WEIBULL ANALYSIS AND LIFE DATA MODELLING

Weibull Analysis utilizes life data distributions governed by the shape parameter (β) to model failure behavior across operating hours:

$\beta < 1$ — Infant Mortality Early life failures driven by manufacturing or assembly anomalies.	$\beta \approx 1$ — Random Failures Constant failure rate driven by unpredictable operational shocks.	$\beta > 1$ — Wear Out Phase Increasing failure rate governed by cumulative fatigue and material ageing.
---	--	--

Lifecycle Timeline: Commissioning to Operational Service and Degradation Ageing

6. COMPARATIVE ANALYSIS: FMEA VS. WEIBULL ANALYSIS

Failure Mode and Effects Analysis (FMEA)	Weibull Life Data Analysis
Identifies potential engineering vulnerabilities	Analyzes empirical field failure time datasets
Evaluates qualitative operational risk severity	Quantifies temporal failure distribution patterns
Leverages Severity, Occurrence, and Detection indices	Applies mathematical shape and scale parameters
Prioritizes engineering design interventions	Predicts time dependent reliability characteristics

Synthesis: FMEA addresses “What can fail?” | Weibull addresses “When does failure occur?”

BORGWARNER TECHNICAL CASE STUDY (CONTINUED)

Root Cause Investigation, Corrective Actions and Reliability Lifecycle Monitoring

7. ACTIONABLE TRANSLATION OF WEIBULL RESULTS

Empirical Failure Data to Weibull Distribution Modeling to Regime Identification to Targeted Engineering Investigation to Root Cause Isolation.

- **Infant Mortality ($\beta < 1$):** Audit upstream machining tolerances, assembly standard operating procedures, and incoming material quality.
- **Random Failures ($\beta \approx 1$):** Investigate environmental stress screening, operating envelope limits, and dynamic load spikes.
- **Wear Out Failures ($\beta > 1$):** Re-evaluate metallurgical fatigue limits, surface treatments, lubrication intervals, and preventative maintenance schedules.

8. ROOT CAUSE ANALYSIS (RCA) FRAMEWORK

Analytical Workflow: Weibull Regime Classification to Ishikawa (Fishbone) Diagramming to 5-Why Interrogation to Physical Root Cause Verification.

Engineering Note: While Weibull modelling identifies temporal failure behaviour, it does not isolate physical root causes. Ishikawa and 5-Why methodologies bridge this gap by tracing symptoms back to fundamental process or design assignable causes.

9. CORRECTIVE AND PREVENTIVE ACTION (CAPA) STRATEGY

Upon isolating root causes, engineering interventions are deployed across manufacturing and design domains:

- Refining CNC machining parameters to reduce dimensional process variation.
- Upgrading material specifications and hardening protocols for enhanced fatigue resistance.
- Implementing automated optical and metrological inspection gates to eliminate escape modes.
- Establishing robust operational guardrails to prevent severe thermal or mechanical overloads.

10. VALIDATION AND ENGINEERING VERIFICATION

Validation Cycle: Engineering Change Implementation to Accelerated Life Testing (ALT) to Statistical Data Comparison to Design Standardisation.

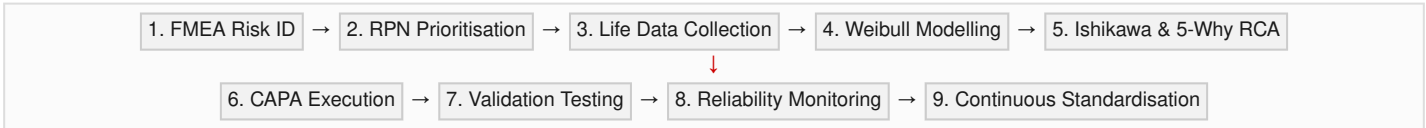
Proposed design and process modifications must be verified through rigorous physical testing. If empirical reliability metrics demonstrate significant improvement, the engineering change is formally integrated into production baselines.

11. CONTINUOUS RELIABILITY MONITORING

Lifecycle Feedback Loop: FMEA Update to Weibull Recalibration to CAPA Verification to Validation to Ongoing Field Monitoring.

Post implementation monitoring ensures that failure rates continue to trend downward and confirms that no secondary failure modes have been inadvertently introduced into the system.

12. COMPLETE QUALITY ENGINEERING WORKFLOW



13. QUALITY ENGINEERING TOOLS UTILISED

Tool Name	Primary Engineering Application in this Case Study
FMEA	Proactive risk identification and calculation of Risk Priority Number (RPN).
Weibull Analysis	Statistical modelling of time-to-failure data and lifecycle regime classification.
Ishikawa (Fishbone) Diagram	Structured brainstorming of potential investigation vectors and root causes.
5-Why Analysis	Iterative interrogation technique to trace symptoms to root assignable causes.
CAPA Framework	Systematic execution of corrective and preventive engineering actions.

14. CONCLUDING ENGINEERING TAKEAWAY

“FMEA establishes a structured baseline for risk prioritisation and failure mode identification, whereas Weibull Analysis decodes temporal failure behaviour across the operational lifecycle. Integrating both methodologies provides a comprehensive framework for proactive quality engineering and reliability enhancement.”

Technical case study independently authored and developed by Jagriti Singh as part of the 100 Days TQM & Quality Engineering Challenge. Not an official BorgWarner project or representation of proprietary BorgWarner enterprise systems, manufacturing data, or internal operational procedures.